Measuring Personalism in Dictatorships: A Latent Trait Approach with Observable Indicators

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Abstract

How can we reliably measure the degree of personalism in dictatorships? Existing regime typologies classify personalist regimes categorically, yet personalism is better understood as a continuous, latent attribute that varies across leaders and over time. We introduce a new cross-national dataset covering 497 authoritarian leader-spells in 123 countries (1946–present), coded on 11 observable "manifest" indicators drawn from publicly available sources—constitutional texts, Wikidata, the Varieties of Democracy project, Wikipedia, and national currency records. We estimate a continuous personalism score for each leader-spell using a two-parameter Item Response Theory (IRT) model. The model achieves strong face validity: leaders widely regarded as highly personalist—including Nkrumah (Ghana)—receive the highest scores, while leaders in institutionalised autocracies score near the bottom. We make the dataset, all code, and the interactive dashboard publicly available to facilitate future research on authoritarian politics.

Keywords: personalism, authoritarianism, measurement, Item Response Theory, dictatorships

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1 Introduction

How can we reliably measure the degree to which dictators have personalized political power? This question matters because personalist rule has profound consequences for governance, conflict, and regime survival (Geddes et al., 2014; Weeks, 2014; Svolik, 2012). Despite its importance, existing measures of personalism suffer from weak measurement validity. Most influential datasets classify regimes into discrete categories—personalist, military, party, or monarchy—using expert judgments that are difficult to replicate and often disagree at the margins (Geddes et al., 2014; Frantz and Ezrow, 2011).

We propose an alternative approach. Rather than classifying regimes categorically, we treat personalism as a continuous, latent trait that can be inferred from observable indicators. Drawing on Treier and Jackman (2008)'s application of Item Response Theory to measuring democracy, we compile 11 binary indicators of personalism from publicly accessible data sources and estimate a latent personalism score (θ) for 497 authoritarian leader-spells across 123 countries.

2 Conceptualising and Measuring Personalism

2.1 Existing Approaches

2.2 A Latent Trait Framework

3 Data

We assemble data from five primary sources, compiling 11 binary indicators for 497 authoritarian leader-spells in 123 countries from 1946 to the present. Leader identification and tenure dates follow Archigos 4.1 (Goemans et al., 2009).

3.1 Indicators of Power Concentration

Our power concentration indicators capture the extent to which the leader has subverted institutional constraints and centralised authority. From the Varieties of Democracy (V-Dem) dataset (Coppedge et al., 2024), we extract dichotomised versions of expert-coded measures: political killings attributed to the state, military domination of the executive, judicial purges, constitutional disregard, and absence of legislative constraints. From the Constitute Project (Elkins et al., 2014), we code whether constitutional term limits are absent and whether the constitution designates the leader as president for life. From Wikidata, we identify leaders who have placed family members in senior government positions.

3.2 Indicators of Personality Cult

Personality cult indicators capture the symbolic elevation of the leader. From Wikidata, we identify places named after leaders, grandiose titles, commissioned monuments, state-mandated birthday holidays, and state hagiographies. From Wikipedia, we flag leaders with documented cults of personality. From national currency records, we code whether the leader's portrait appears on banknotes during their tenure. From the Constitute Project, we code whether oaths of loyalty are directed to the leader personally rather than to the constitution.

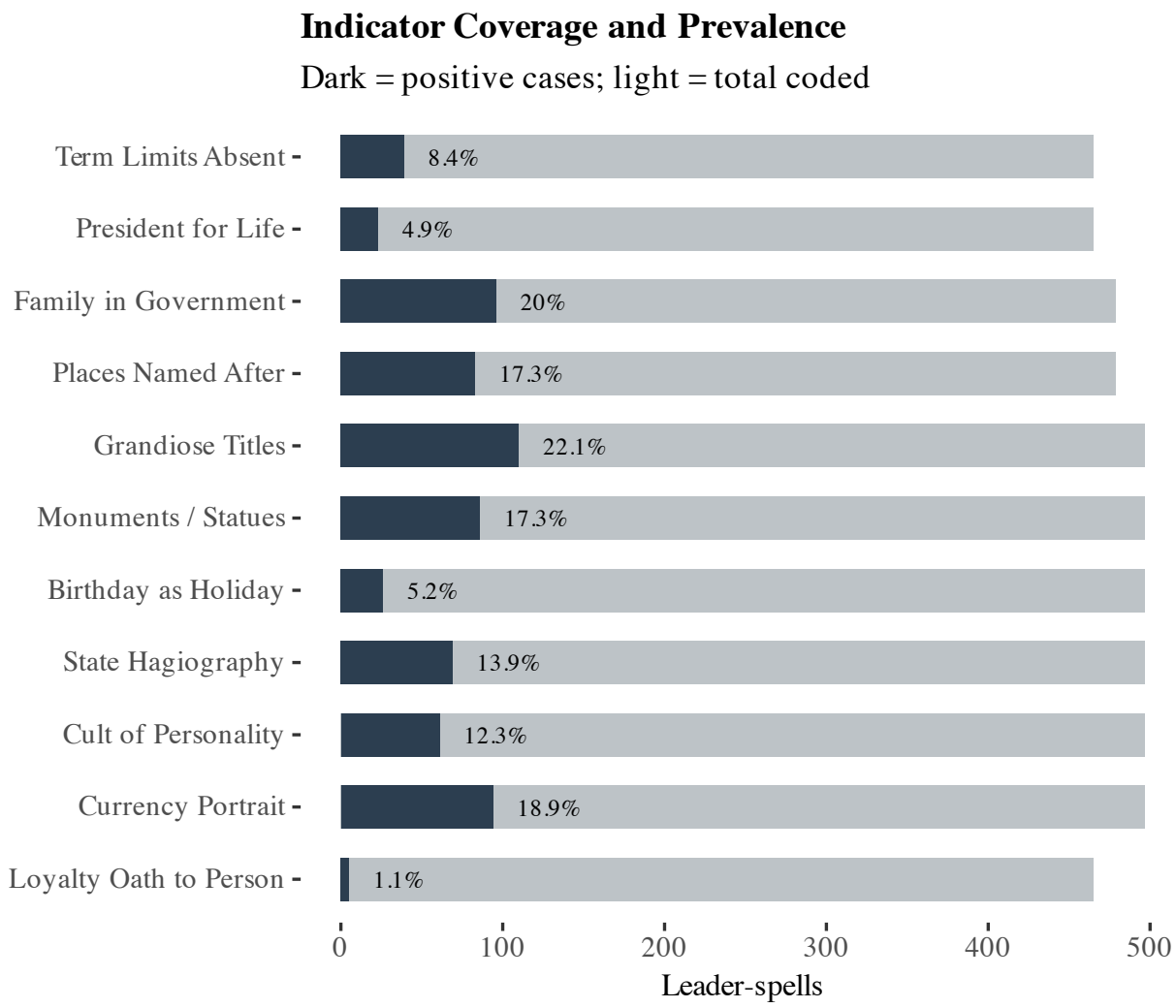

Figure 1: Coverage and prevalence of the 11 manifest indicators. Dark bars indicate positive codings; light bars show total coded cases.

4 Methods

We estimate a unidimensional two-parameter logistic (2PL) Item Response Theory model (de Ayala, 2009; Embretson and Reise, 2000). For leader-spell i and indicator j:

$$P(Y_{ij} = 1 \mid \theta_i) = \frac{1}{1 + \exp\big(-a_j(\theta_i - b_j)\big)} \tag{1}$$

where θ_i is the latent personalism score, a_j is the discrimination parameter (how strongly

indicator j relates to personalism), and b_j is the difficulty parameter (the level of personalism at which the indicator becomes more likely than not). Parameters are estimated via marginal maximum likelihood with the EM algorithm using the `mirt` package in R (Chalmers, 2012). Missing values are handled natively under the assumption that they are missing at random.

Latent scores are estimated using Expected A Posteriori (EAP) estimation with associated standard errors, following Treier and Jackman (2008).

5 Results

5.1 Item Parameters

Figure 2 displays the estimated discrimination and difficulty parameters for all 11 indicators.

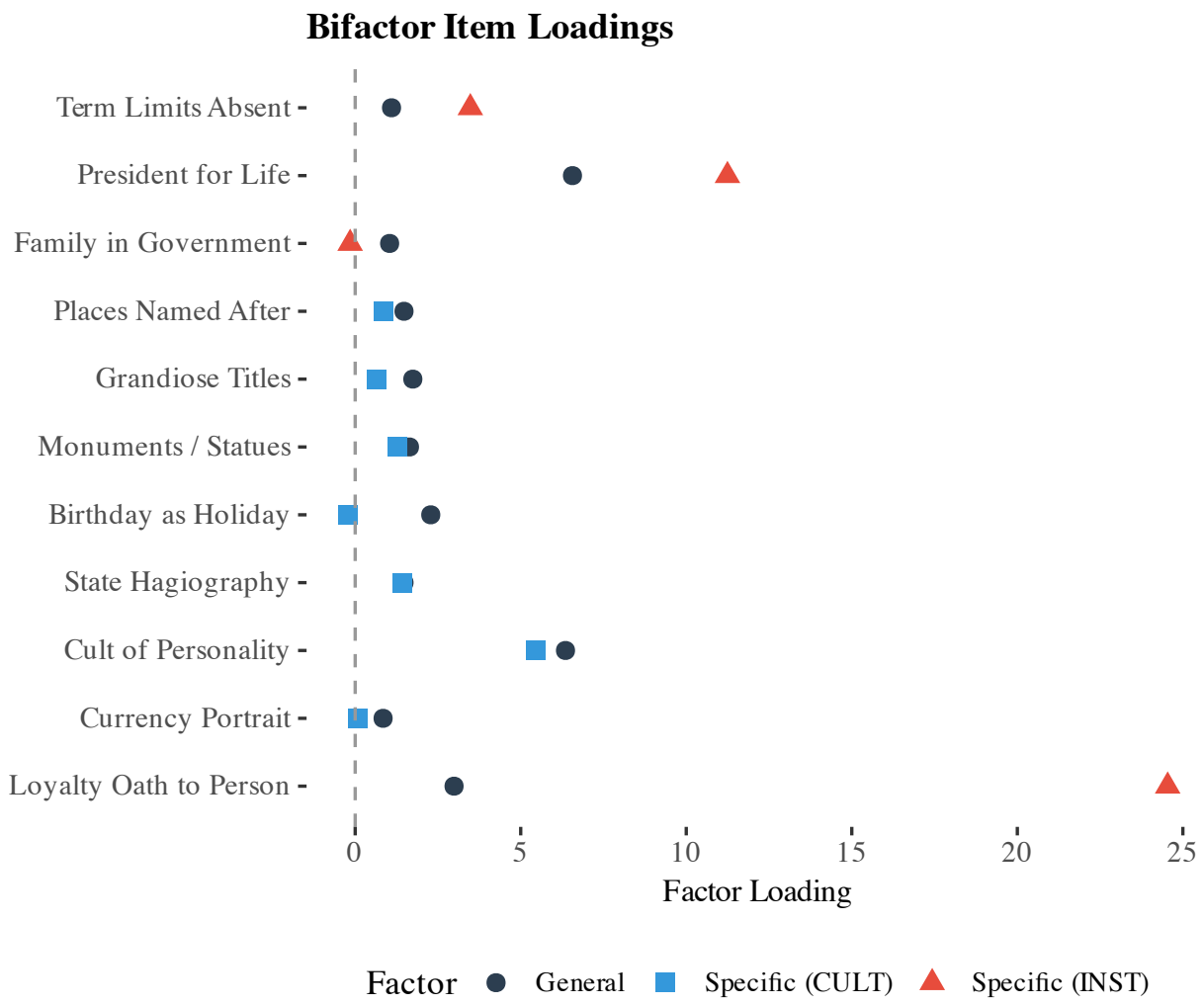

Figure 2: Estimated item parameters from the 2PL IRT model. Discrimination (a) reflects how strongly each indicator relates to the latent personalism trait; difficulty (b) reflects the level of personalism required for the indicator to be more likely present than absent.

5.2 Personalism Scores

The distribution of estimated personalism scores ($\hat{\theta}$) is shown in Figure 3. Scores have a mean of -0.000 (SD = 0.712), ranging from -0.646 to 2.403.

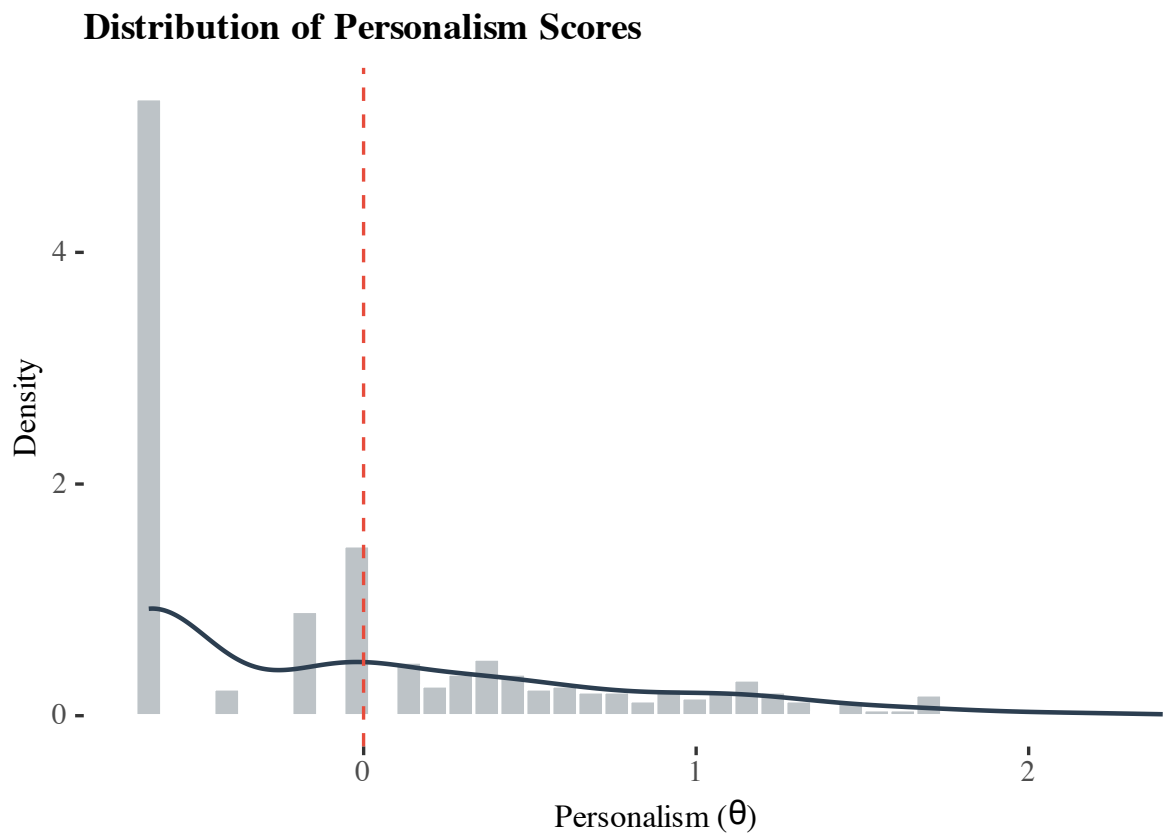

Figure 3: Distribution of estimated personalism scores ($\hat{\theta}$) across 497 leader-spells. The dashed line indicates the mean.

5.3 Face Validity

Figure 4 presents the 25 leaders with the highest estimated personalism scores. The ranking provides strong face validity: leaders widely regarded as highly personalist—such as Nkrumah (Ghana)—occupy the top positions.

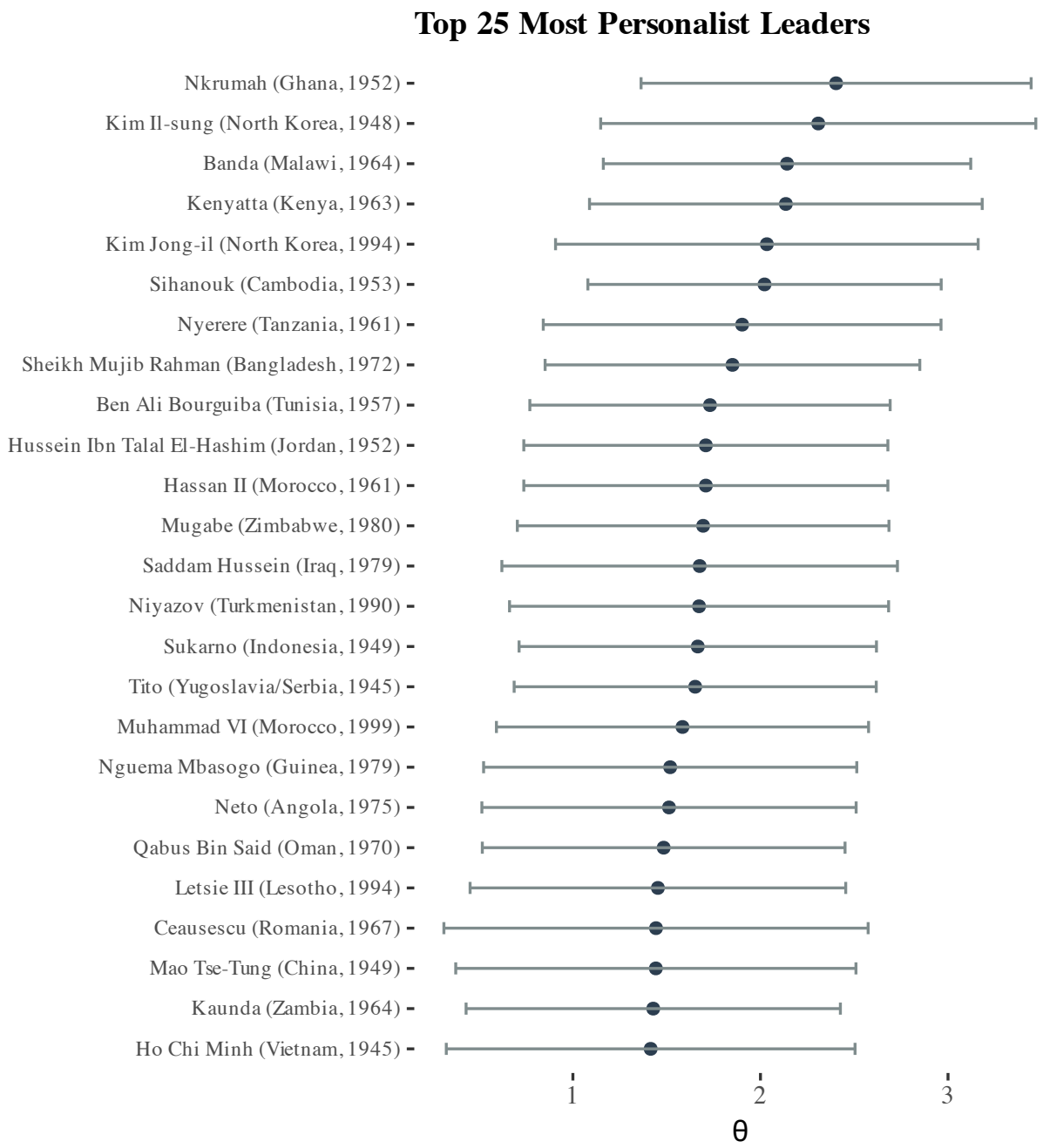

Figure 4: The 25 leaders with the highest estimated personalism scores, with 95% confidence intervals.

5.4 Test Information

Figure 5 displays the test information function. The measure provides the most information—and therefore the most precise estimates—in the range of θ where personalism transitions

from moderate to high levels.

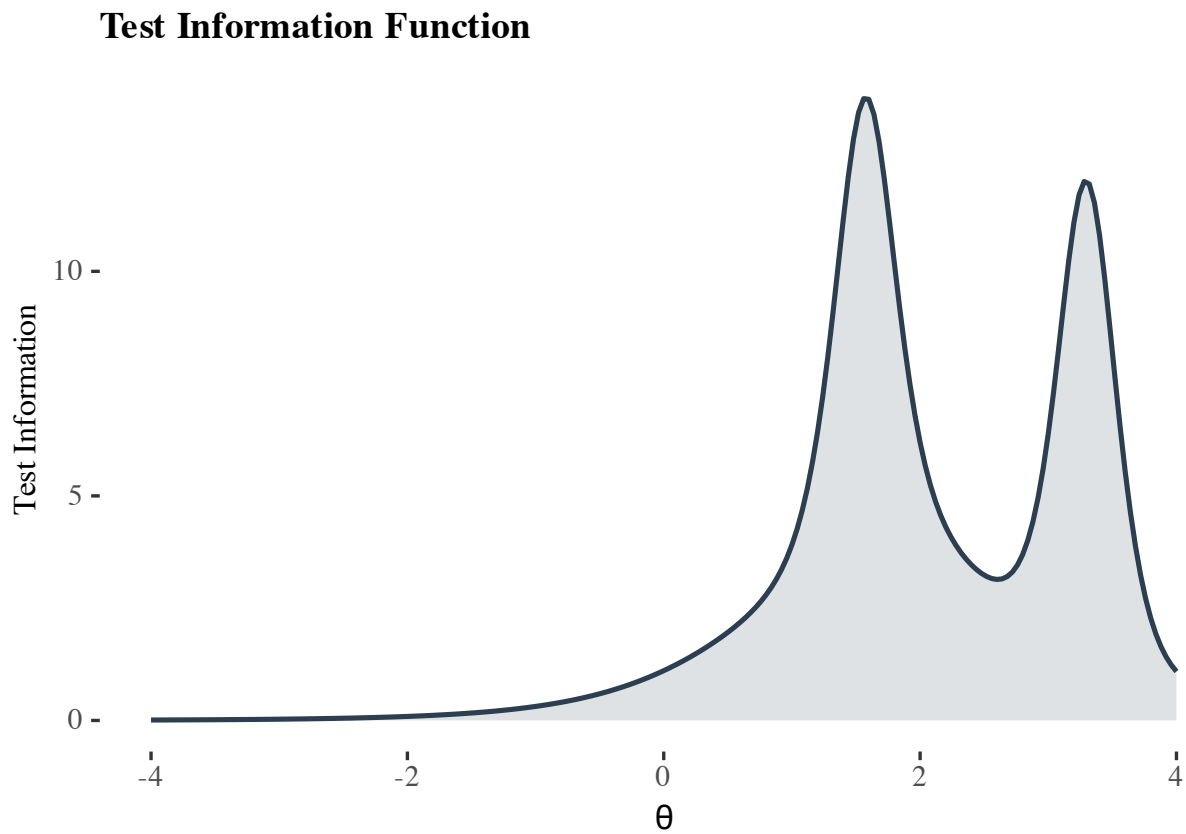

Figure 5: Test information function. Higher values indicate greater measurement precision at that level of θ.

5.5 Model Fit

5.6 Comparison with GWF Classifications

As an external validation exercise, we compare our continuous personalism scores with the categorical regime classifications from Geddes et al. (2014). Of 497 leader-spells, 351 can be matched to GWF regime-years by COW code and temporal overlap; 146 remain unmatched, primarily because they govern after GWF coverage ends in 2010. Among matched leaders, 156 govern regimes classified as containing a personal element ("personal," "party-personal," "military-personal," or "party-personal-military") and 195 govern non-personal regimes.

Figure 6 displays the distribution of $\hat{\theta}$ by GWF binary classification. Leaders in personal

regimes score marginally higher ($\bar{\theta} = 0.111$, SD = 0.746) than those in non-personal regimes ($\bar{\theta} = 0.081$, SD = 0.756), but the difference is small and not statistically distinguishable from zero (point-biserial $r = 0.019$, $p = 0.7168$; Cohen's $d = 0.039$; Welch t = -0.364, $p = 0.7164$).

The weak association is substantively informative. GWF classifies *regimes*, while our measure captures *individual-level* personalism across multiple observable dimensions, including personality cult indicators that GWF does not directly code. A leader may exhibit strong personalist tendencies—accumulating titles, naming places, commissioning statues—within a regime that GWF categorises as party-based or military. Conversely, some leaders in GWF-designated personal regimes may inherit personalist institutional structures without cultivating personality cults. This conceptual divergence supports our argument that personalism is better understood as a continuous, multidimensional trait than as a categorical regime property.

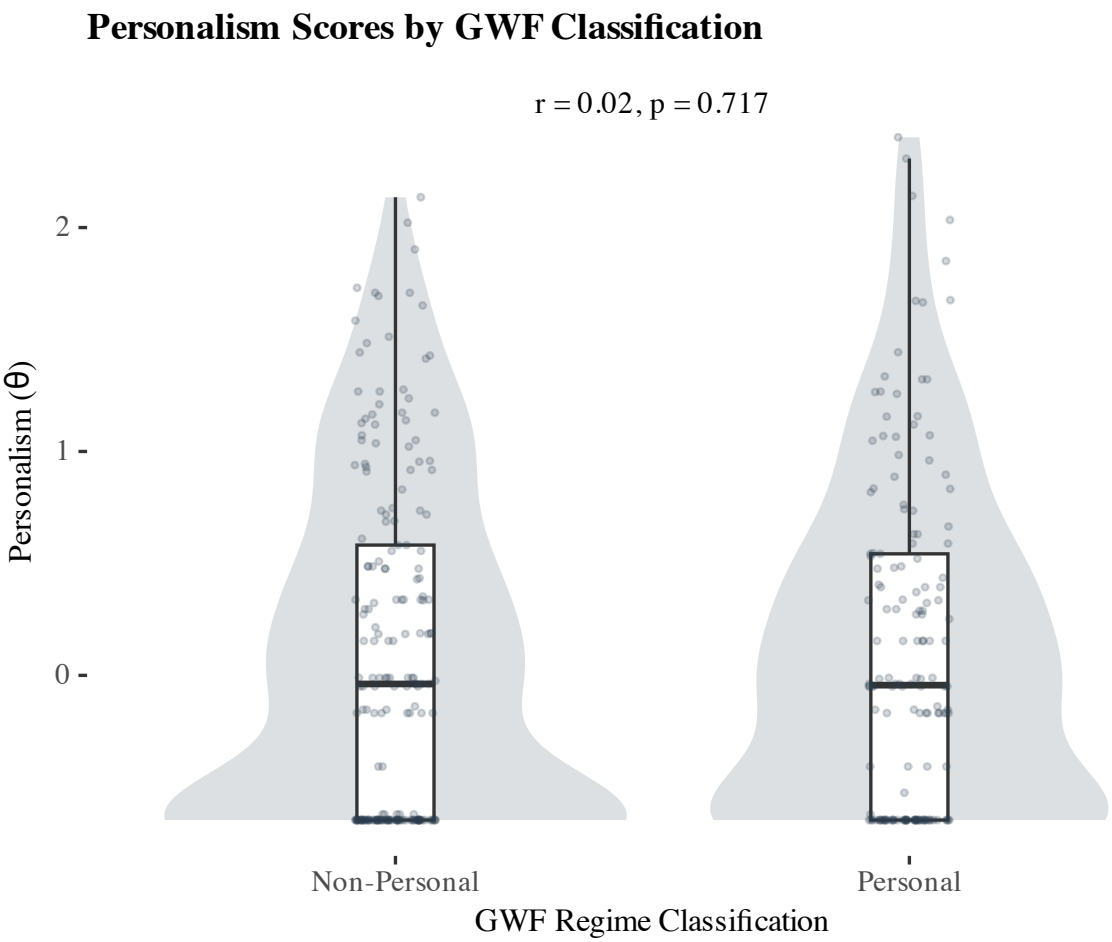

Figure 6: Distribution of personalism scores ($\hat{\theta}$) by GWF regime classification. "Personal" includes all GWF types containing a personal element. Boxes show interquartile ranges; violin contours show kernel density estimates.

6 Discussion

6.1 Limitations

6.2 Extensions

7 Conclusion

Data Availability. We will make all data and code used to generate our results available at a figshare repository at the time of publication.

Acknowledgments. [TBD]

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A Item Characteristic Curves

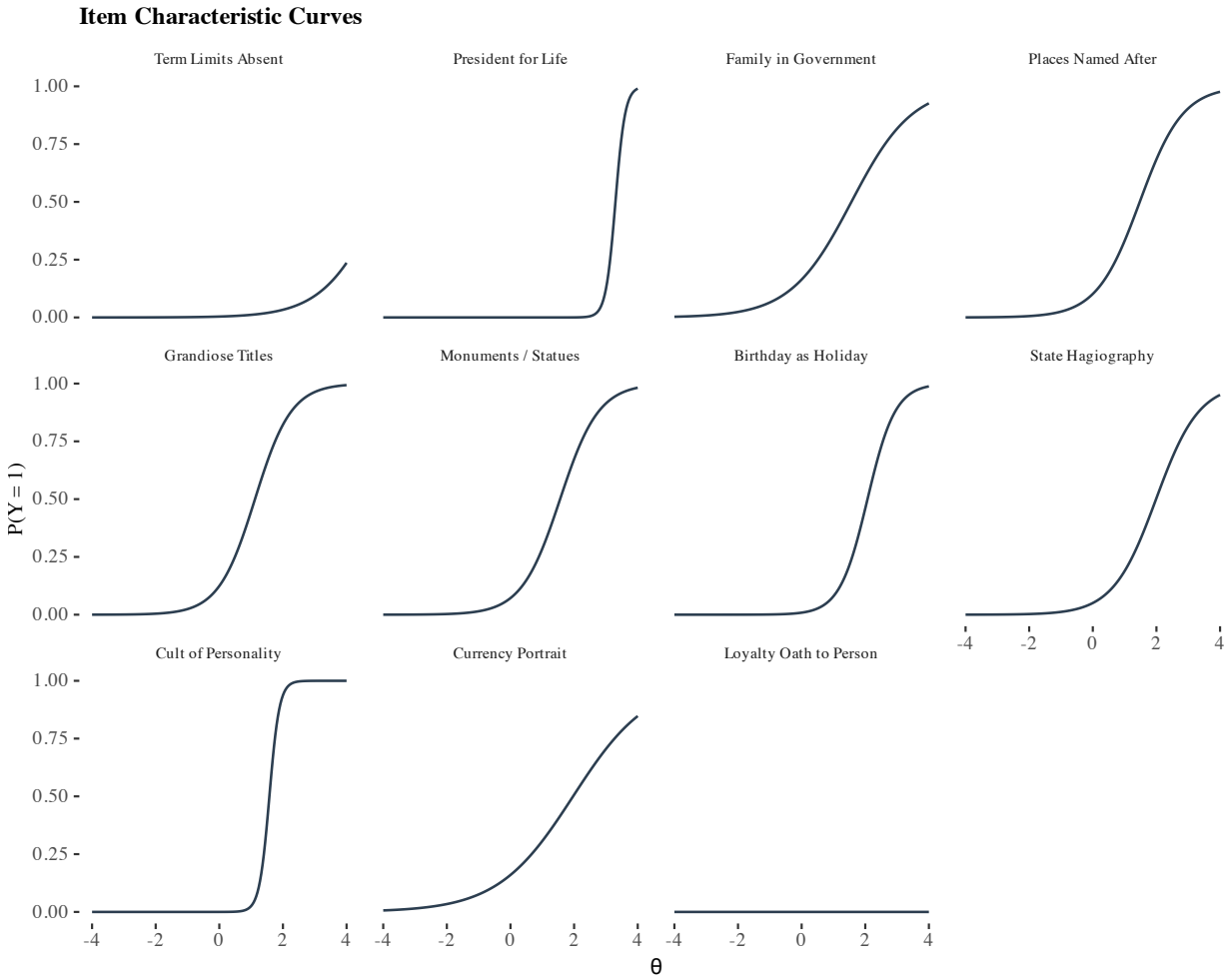

Figure 7: Item Characteristic Curves (ICCs) for all 11 indicators, showing the probability of a positive coding as a function of the latent personalism score θ.

B Bottom-Ranked Leaders

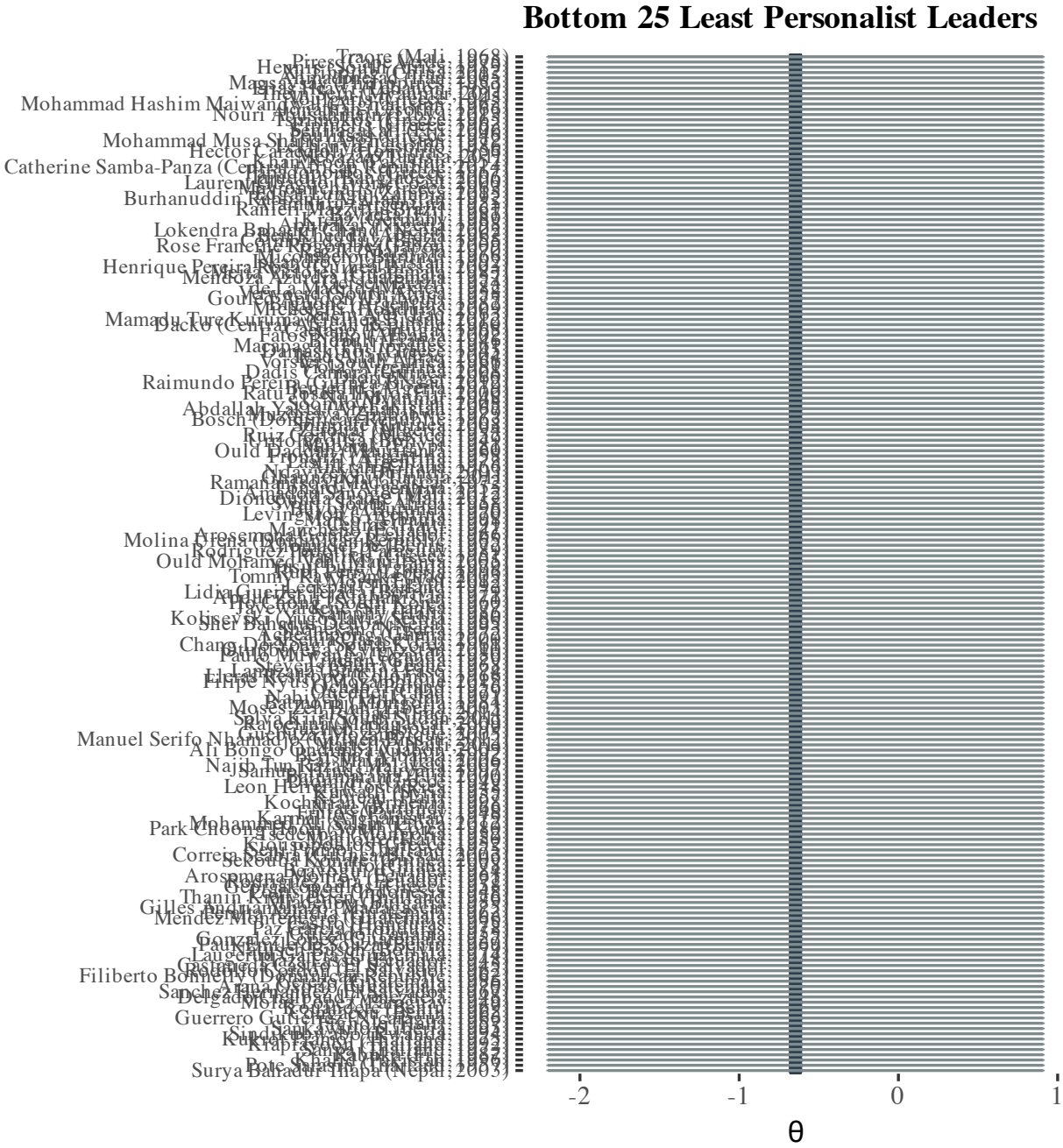

Figure 8: The 25 leaders with the lowest estimated personalism scores, with 95% confidence intervals.